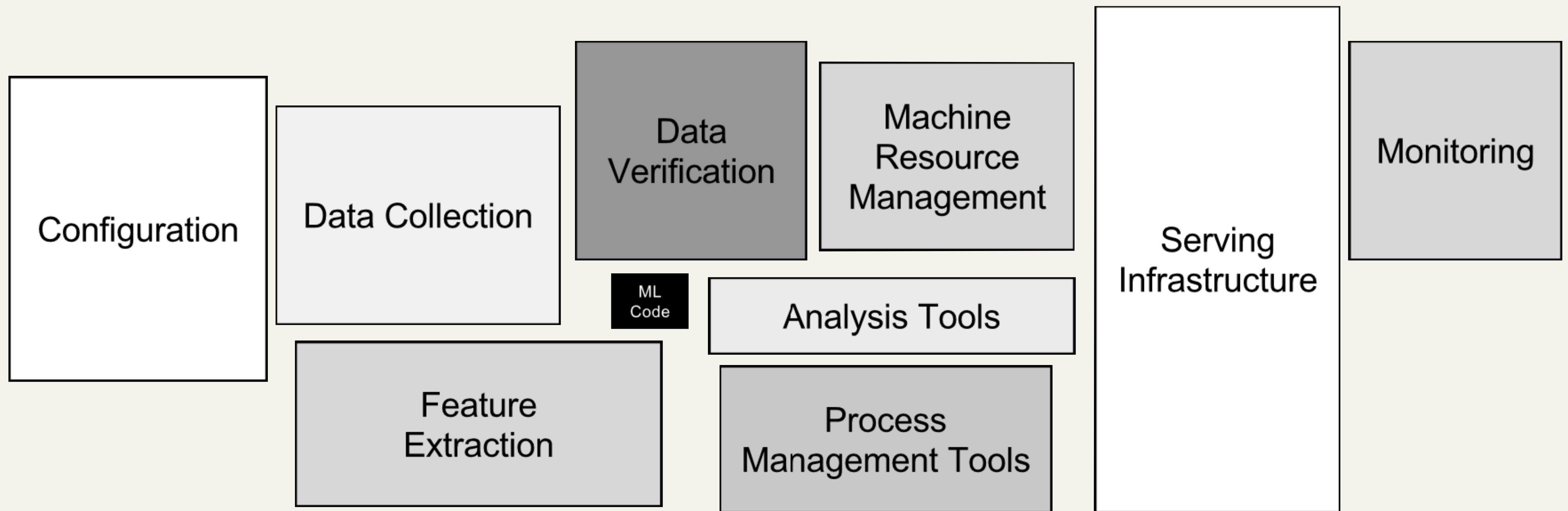


From DevOps to MLOps to LLMOps

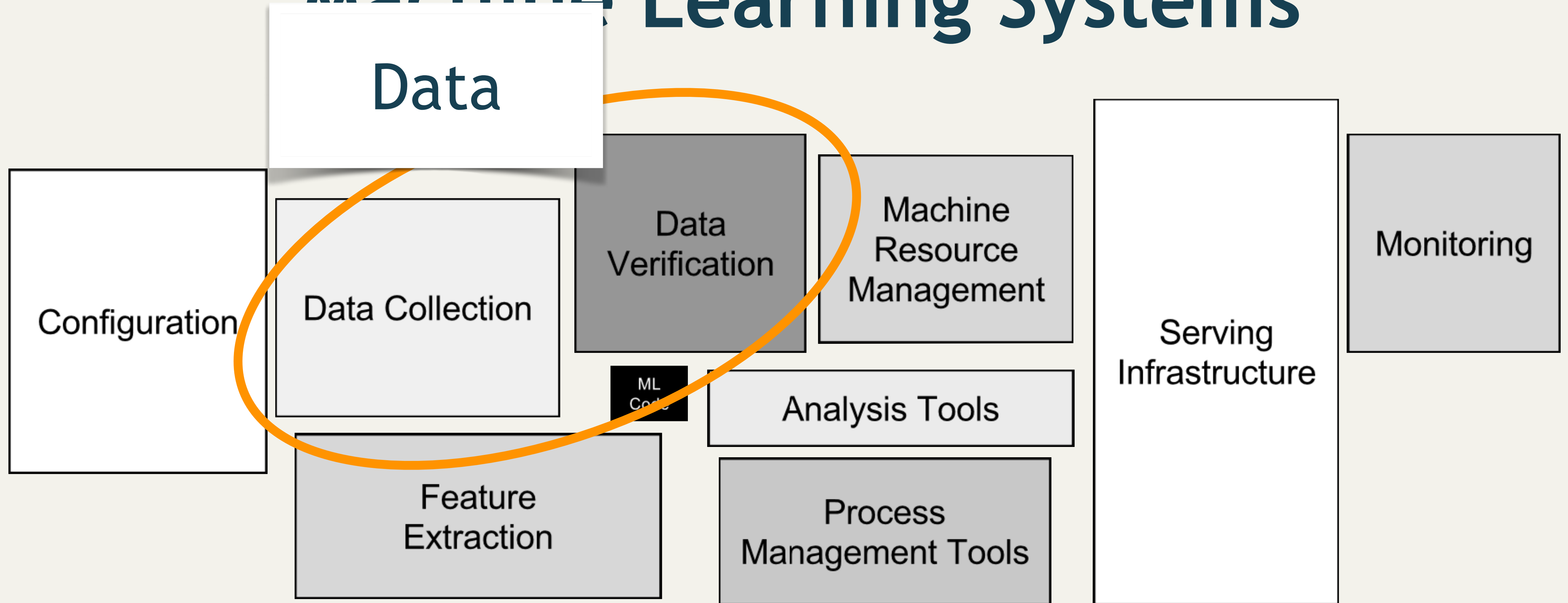
Dimitrios Verraros, October 2026



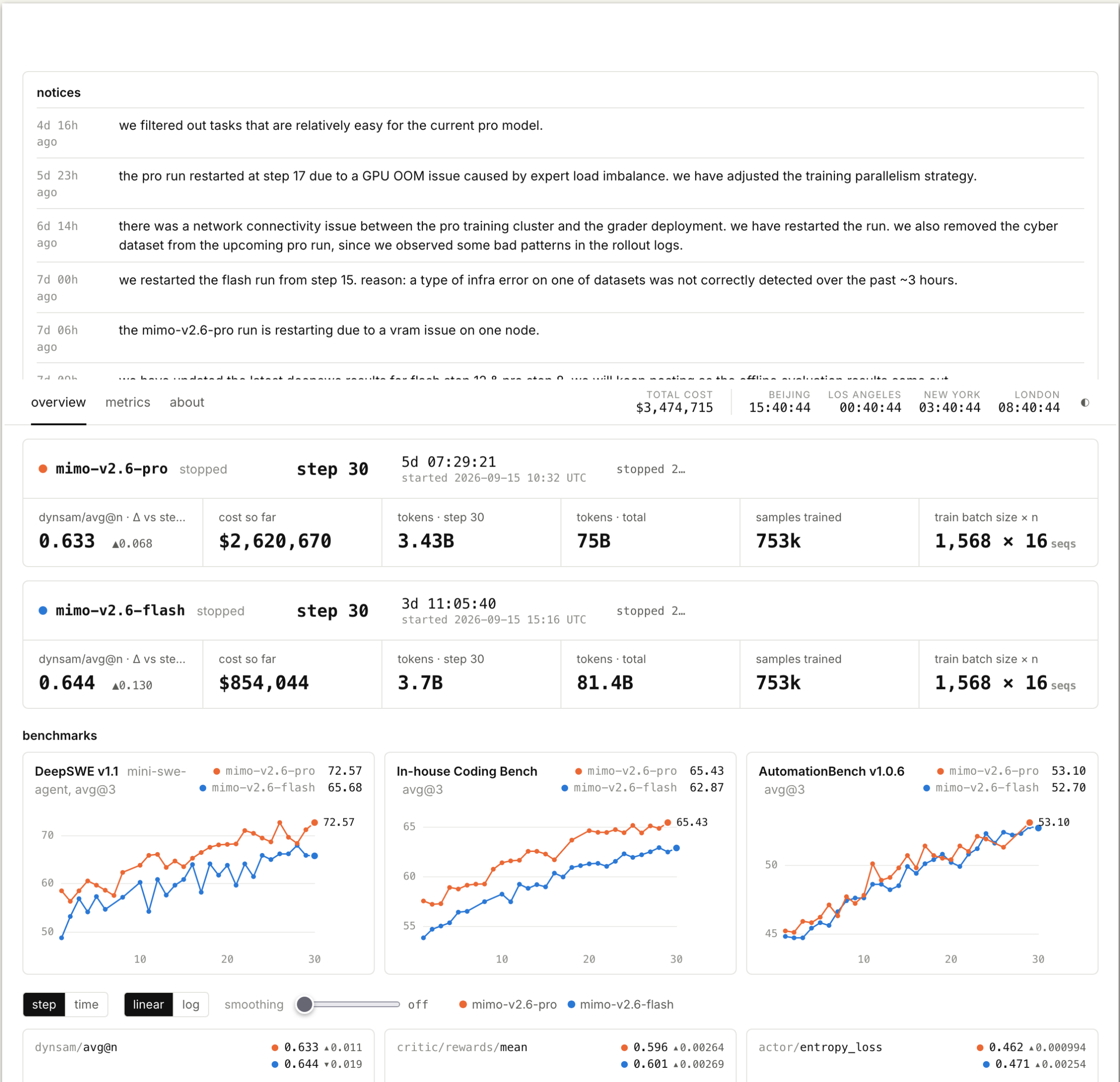
Machine Learning Systems



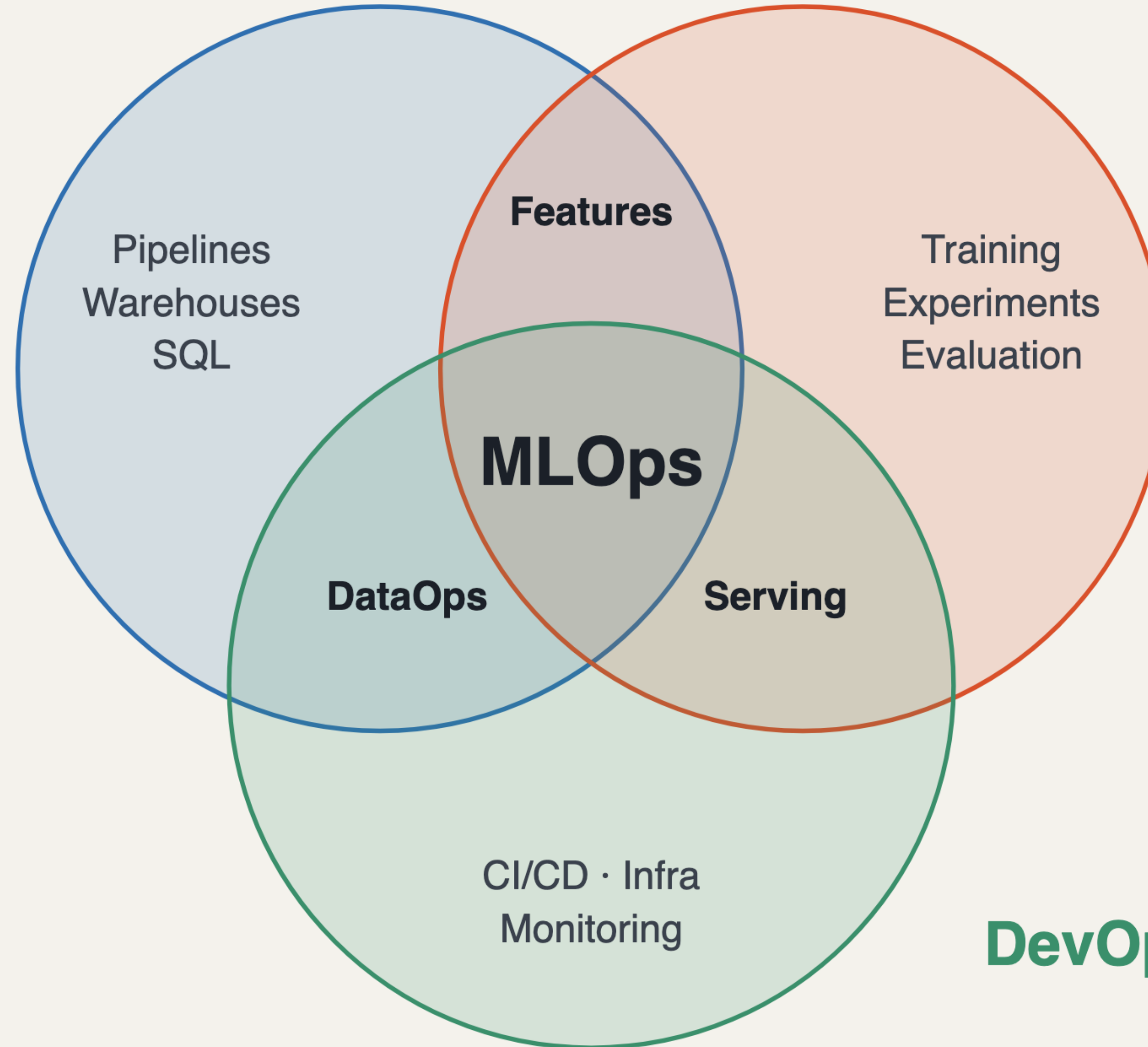
Machine Learning Systems



Mimo Live Training Dashboard



**Data
Engineering**



**Machine
Learning**

DevOps

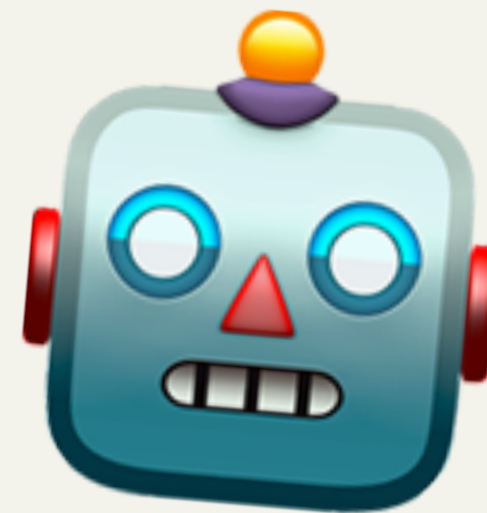
DevOps

Software
Development



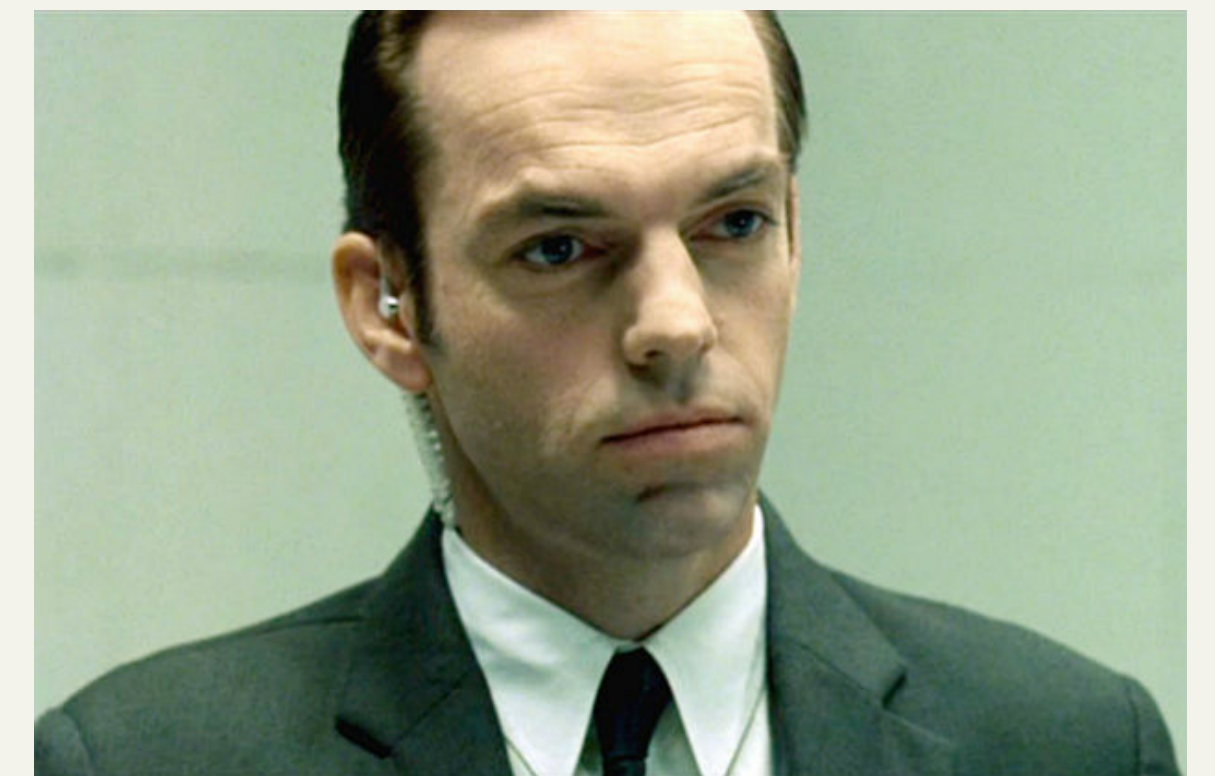
MLOps

Machine
Learning



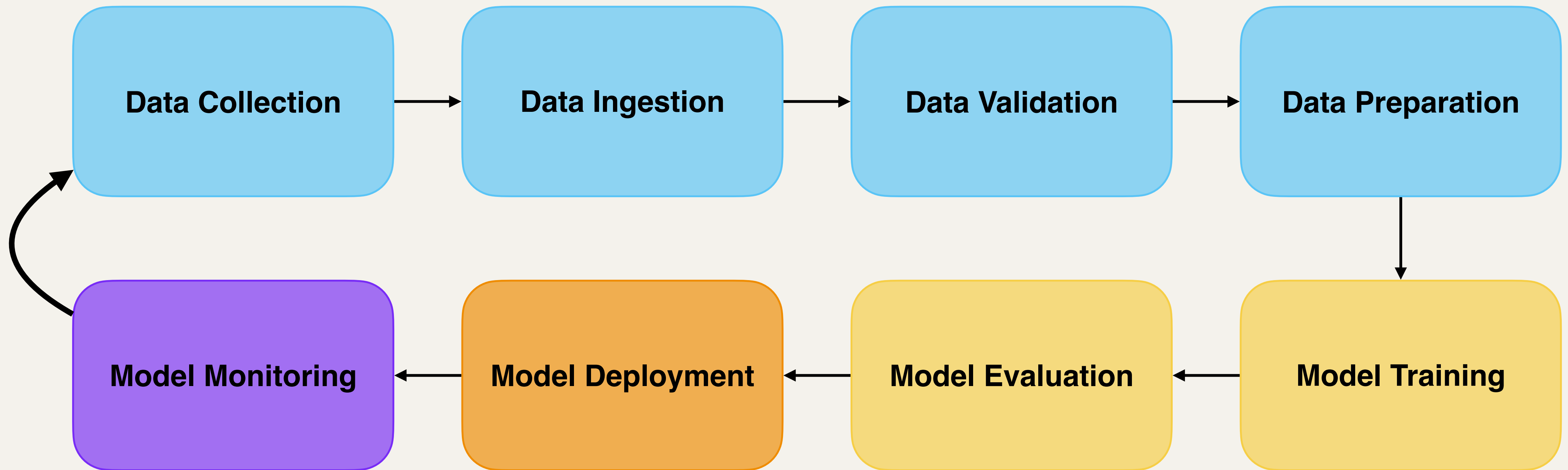
LLMOps

Agents



AI Ops?

MLOps



Collection

Ingestion

Validation

Preparation

Training

Evaluation

Deployment

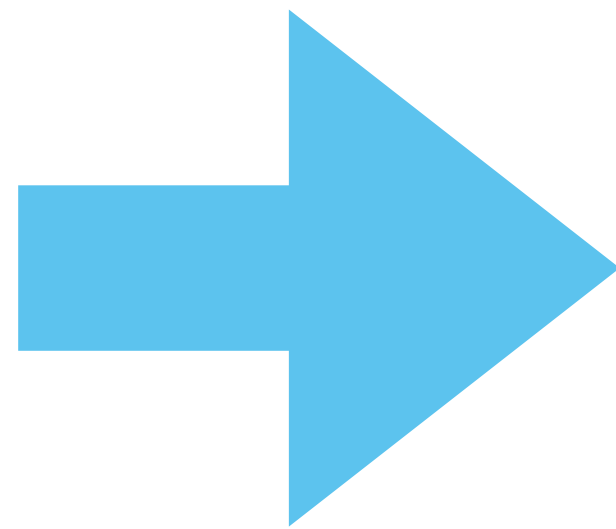
Monitoring

Web & text

Images & Video

Sensors

Logs



Object Store

Common tools: Kafka, Scrapy, S3

Collection

Ingestion

Validation

Preparation

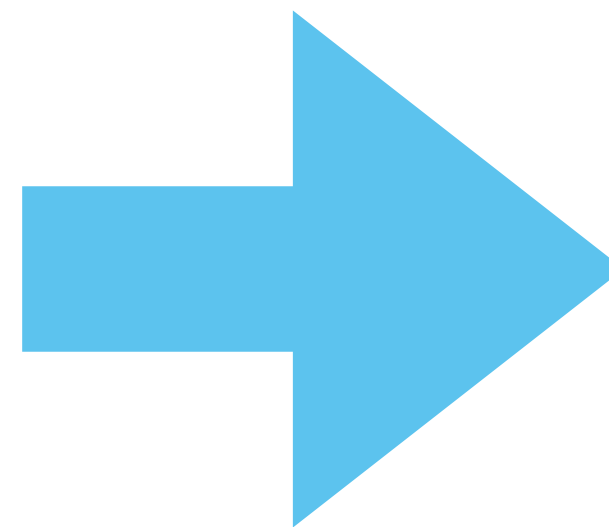
Training

Evaluation

Deployment

Monitoring

Raw Data



Data Lake
Warehouse

Common tools: Spark, Airflow, SnowFlake, Postgres

Collection

Ingestion

Validation

Preparation

Training

Evaluation

Deployment

Monitoring

Data Review



✓ Clean data

✗ Flag

Common tools: Great Expectations, Pandera

Collection

Ingestion

Validation

Preparation

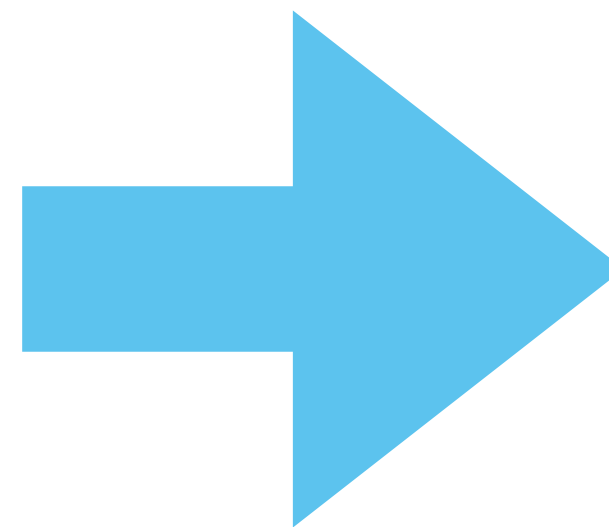
Training

Evaluation

Deployment

Monitoring

DB Tables
Labels



Data Sets
Transformations

Common tools: Spark, dbt, Parquet, Feather

Collection

Ingestion

Validation

Preparation

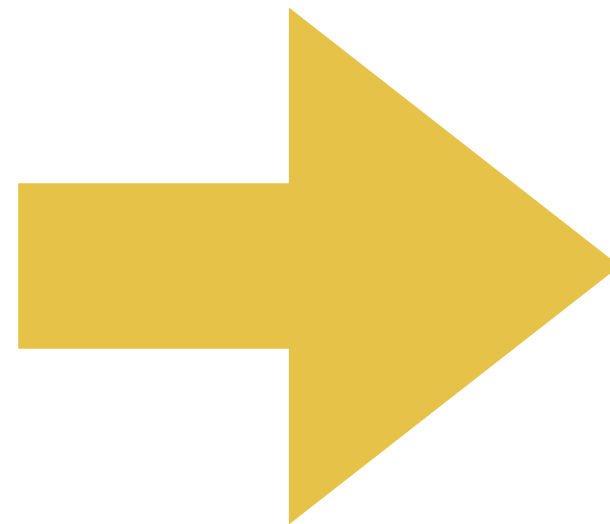
Training

Evaluation

Deployment

Monitoring

Training set
Code
Config



Model
Experiment Log

Common tools: **PyTorch**, **Tensorflow**, **MLFlow**, **Ray**

Collection

Ingestion

Validation

Preparation

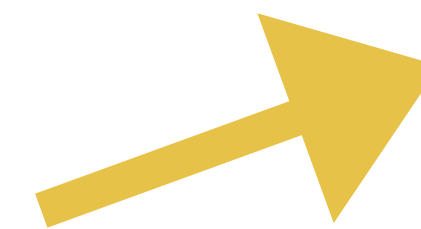
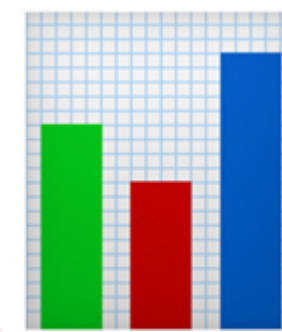
Training

Evaluation

Deployment

Monitoring

Models
Test set



✓ Approve



✗ Reject

Common tools: **Weights & Biases, MLFlow Registry**

Collection

Ingestion

Validation

Preparation

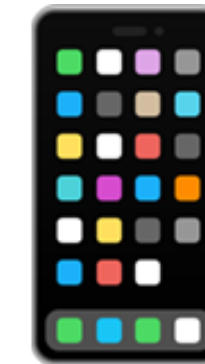
Training

Evaluation

Deployment

Monitoring

Model Delivery



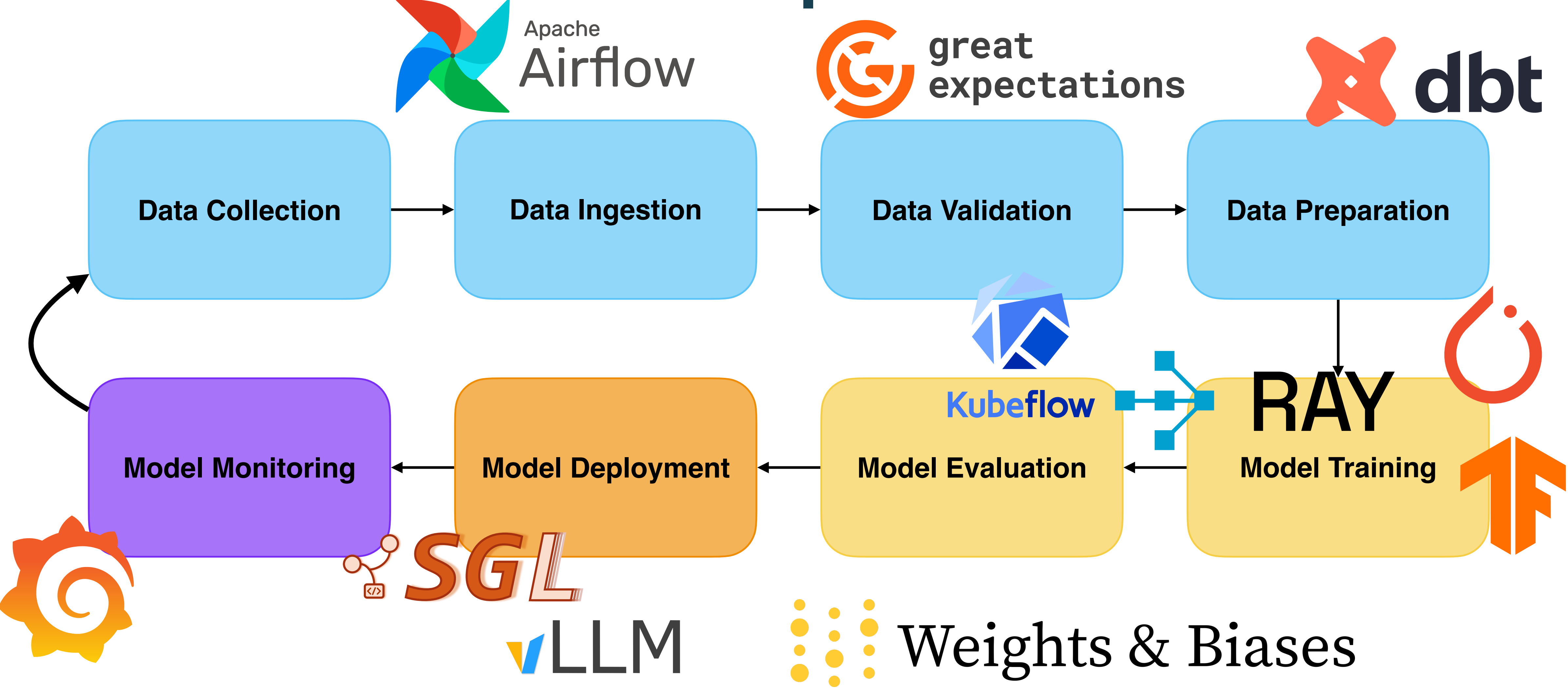
Common tools: KServe, kagent, vLLM



Latency
Throughput
Error Rates

Common tools: Prometheus, Grafana

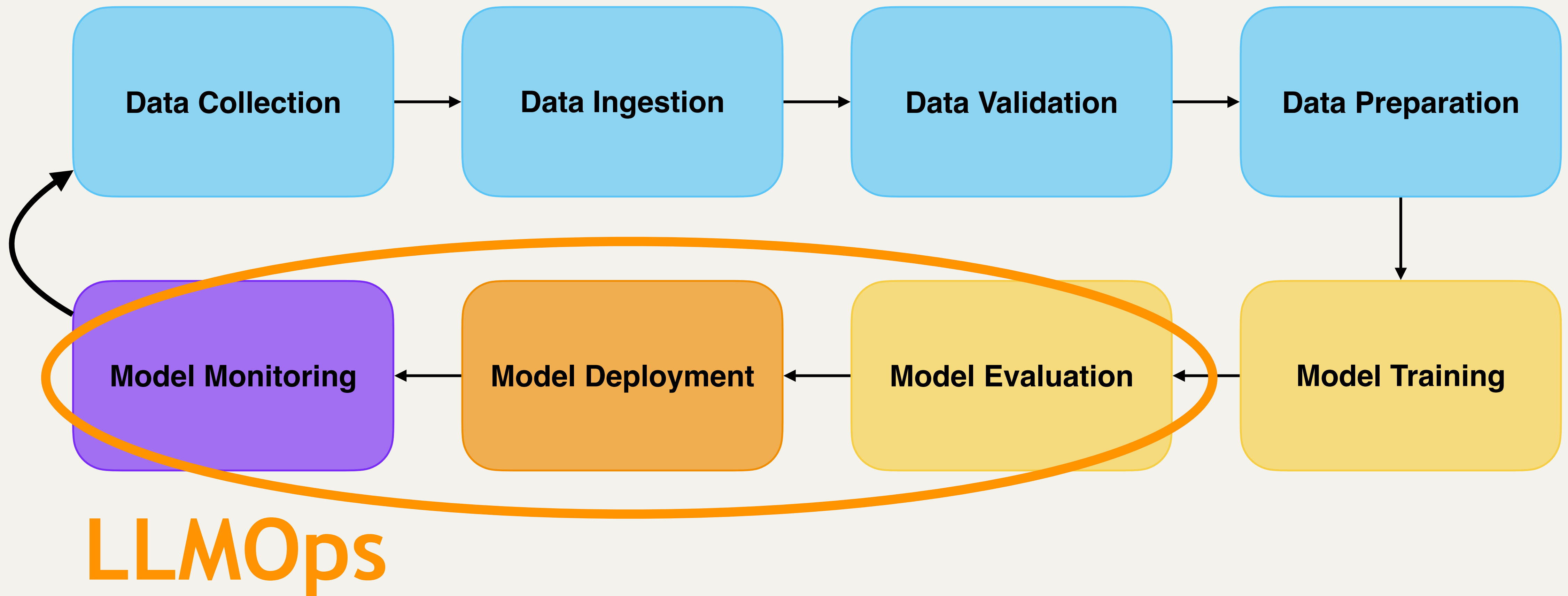
MLOps



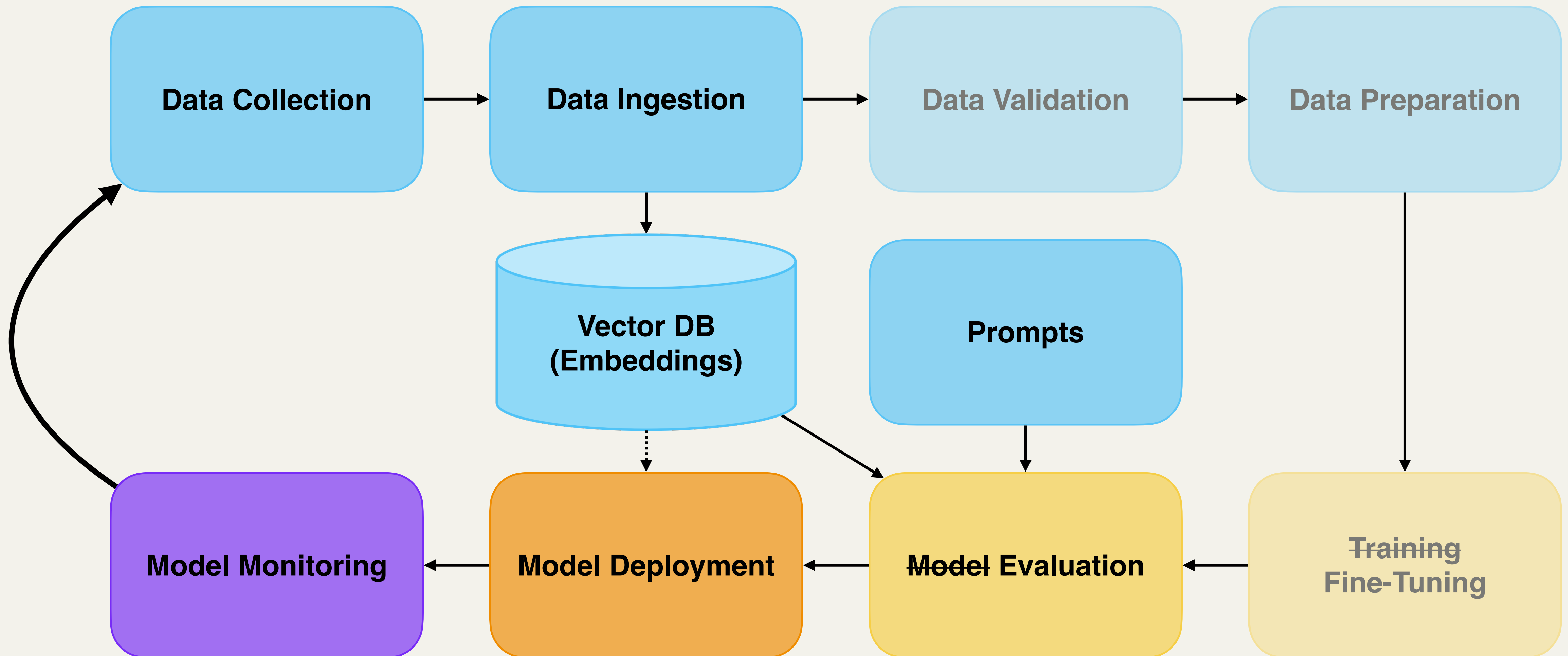
DevOps to MLOps

Area	You already know	You need to learn
Data	DB Ops, Storage	SQL, Parquet, S3, Datasets
Code	Bash, YAML, Terraform	Python and Ecosystem
Compute	Kubernetes, Swarm, EC2	GPUs
Pipelines	CI/CD	Data and Training Pipelines
Artifacts	Images, SBOM	Models and the data that made them (Data Lineage)
Monitoring	Uptime, latency, errors	Accuracy, Drift
Collaboration	Developers	ML Engineers

What about LLMOps



What about LLMOps



DevOps to LLMOps

Area	DevOps	MLOps	LLMOps
Data	DB Ops, Storage	SQL, Parquet, S3, Datasets	Vector Databases
Code	Bash, YAML, Terraform	Python and Ecosystem	Prompts ;)
Compute	Kubernetes, Swarm, EC2	GPUs	
Pipelines	CI/CD	Data and Training Pipelines	RAG
Artifacts	Images, SBOM	Models	Prompts, Vector Indexes
Monitoring	Uptime, latency, errors	Accuracy, Drift	Throughput, Quality, Cost
Collaboration	Developers	ML Engineers	Product Engineers

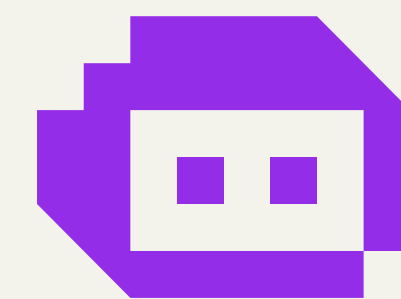
Demos

MLOps

ryaki

Train your LLM

LLMOps



kagent